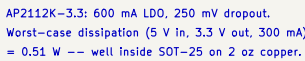


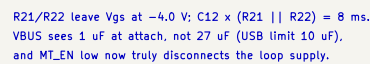
USB-C INPUT & ESD



STATUS



SOFT-START LOAD SWITCH



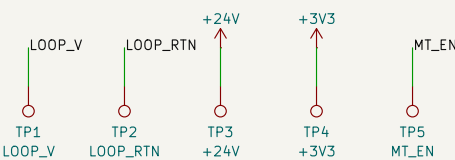
BOOST CONTROLLER



Load is one 2-wire transmitter at 24 mA max, so the converter runs at roughly 30 mA / 0.72 W out.

D2 (SS34) and C11 are rated 40 V / 50 V for margin over the 24 V rail; U5's SW pin is the tight one, 30 V.

24 V OUTPUT & FEEDBACK



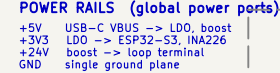
SUPPORT CIRCUITRY



IO0 (27) – Internal pull-up, held high by SW2 idle.
IO46 (16) – Internal pull-down; leave floating (NC).
IO45 (26) – VDD_SPI select; must stay low on -N4.
Do not add a pull-up here.
IO3 (15) – no internal pull; JTAG source select only
floating is safe with no JTAG header.

USB: the S3's USB-Serial/JTAG block handles reset and bootloader entry over native USB, so no DTR/RTS transistor circuit is needed.

INA226 CURRENT MONITOR



SECTION LINKS (net labels)

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USB_P / USB_N   USB-C -> ESP32-S3
I2C_SDA / I2C_SCL ESP32-S3 -> INA226

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SHUNT SIZING

R12 = 3.32 ohm: the INA226's ± 81.92 mV full scale then reaches 24.67 mA, so a sensor driving 23.5 mA over-range still reads instead of clipping.
24 mA $\times$ 3.32 = 79.7 mV (97% of range); LSB 2.5 μ V $\rightarrow$ 0.75 μ A. Dissipation at 24 mA = 1.9 mW (0805).

R13/R14 + C9 form a 10 ohm / 100 nF differential input filter (f_c - 80 kHz). Keep the R12 ground end and R14 on the same node -- Kelvin return.

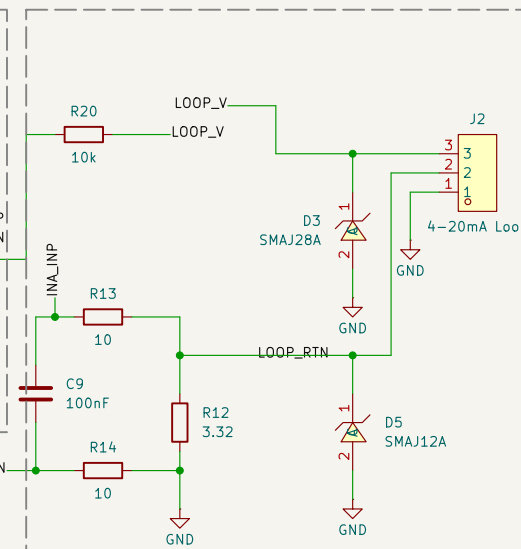
D3/D5 clamp each field wire to ground: 45 V / 20 V
R20 holds D3's clamp into U3's 40 V bus pin to
0.5 mA of its 5 mA rating (costs 1.19% of gain).

LIMITE

$I_{lim} = V_{be}(Q2) / R16: 35 / 33 / 27 \text{ mA at } -20 / 25 / 85 \text{ C, so } 24 \text{ mA always reads.}$
 Dead short: $24 \text{ V} \times 35 \text{ mA} = 0.85 \text{ W in } Q1,$
 hence SOT-223. R16 + Q1 + D4 cost 1.2 V:
 22.8 V at J2 with 20 mA flowing.

J2: 1 = GND, 2 = mA in, 3 = +24 V out.
2-wire sensor on 3-2; sourcing sensor on 2-1

SHUNT, FILTER & FIELD TERMINAL



CURRENT LIMITER

